

A 24-PERMUTATION HAMILTONIAN CYCLE OF $P(4)$ THAT DOES NOT EXTEND TO A U-CYCLE

ABSTRACT. Let $P(n)$ be the digraph of overlapping permutations: its vertices are the $n!$ permutations of $\{1, \dots, n\}$, with an arc $x_1 \cdots x_n \rightarrow y_1 \cdots y_n$ when $x_2 \cdots x_n$ and $y_1 \cdots y_{n-1}$ are order-isomorphic. Chung, Diaconis and Graham introduced a procedure turning a Hamiltonian cycle of $P(n)$ into a universal cycle for n -permutations, provided the strict inequalities the cycle forces on the letters have no directed cycle. Kitaev and Qiu report that the authors of that paper believe this to be the case for *any* Hamiltonian cycle of $P(n)$, and in 2026 restate the belief as a conjecture. We exhibit a directed Hamiltonian cycle of $P(4)$ whose forced inequalities contain the directed cycle $u_2 < u_5 < u_7 < u_9 < u_{12} < u_{14} < u_{17} < u_{20} < u_{23} < u_2$. It therefore admits no linear extension and does not extend to a u-cycle, so the universally quantified statement as those two papers report it fails already at $n = 4$. The proof is a 24-entry table, 24 arc checks and nine readings of a window, and needs no computation. We did not obtain the 1992 paper itself and make no claim about what it states: what is refuted is the statement as reported in 2024 and conjectured in 2026. The existence of u-cycles for n -permutations, and Hurlbert's construction to which Chung, Diaconis and Graham refer, are untouched.

1. THE STATEMENT THAT IS REFUTED

For a word w of distinct letters, $\text{st}(w)$ denotes its standardisation: the permutation of $\{1, \dots, |w|\}$ order-isomorphic to w . The digraph is defined by Kitaev and Qiu [6] as follows; we quote the $\text{\LaTeX}$ source of their e-print `arXiv:2408.05984v1` verbatim, inserting only line breaks for width.

the vertex set of $P(n)$ is the set of all $n!$ permutations of $\{1, 2, \dots, n\}$, and there is an edge $x_1 x_2 \cdots x_n \rightarrow y_1 y_2 \cdots y_n$ if and only if, for $2 \leq i \leq n$, $x_i < x_j$ if and only if $y_{i-1} < y_{j-1}$.

Equivalently, $p \rightarrow q$ is an arc of $P(n)$ if and only if $\text{st}(p_2 \cdots p_n) = \text{st}(q_1 \cdots q_{n-1})$.

Given a directed Hamiltonian cycle $C = (p_0, \dots, p_{N-1})$ of $P(n)$, where $N = n!$, the Chung–Diaconis–Graham procedure [2] introduces N undetermined letters $u_0, \dots, u_{N-1}$ placed on a circle and requires, for every i , that the *window*

$$(u_i, u_{i+1}, \dots, u_{i+n-1}), \quad \text{indices mod } N,$$

be order-isomorphic to p_i . Any two positions lying in a common window therefore have their relative order forced. Write $D(C)$ for the digraph on $\{0, \dots, N-1\}$ whose arcs $a \rightarrow b$ are the forced strict inequalities $u_a < u_b$. If $D(C)$ is acyclic then any linear extension of

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it gives a u-cycle for n -permutations; if $D(C)$ has a directed cycle then no assignment of letters realises all N windows and C yields nothing.

The belief we refute is reported later in the same file. Quoted exactly, including the source’s own grammatical slip “believe that this to be the case”:

Even though obtaining a Hamiltonian cycle $\mathcal{C}$ in $\mathcal{P}(n)$ is straightforward, arguing that the implied ordering on the values is in fact a partial order, i.e. $\mathcal{C}$ has no cycles, is a difficult task. While the authors of [CDG1992] believe that this to be the case for *any* Hamiltonian cycle in $\mathcal{P}(n)$, they refer to [Hurlbert] where a particular construction of $\mathcal{C}$ is offered that guarantees the ordering be a partial order.

The load-bearing word is the quantifier *any*. The same two authors restate it as an open conjecture in 2026 [7] (arXiv:2603.01005v1); we quote that sentence *together with its antecedent*, because the antecedent is what fixes the class quantified over.

Any Eulerian cycle in a graph formed by clusters can be extended to a Hamiltonian cycle in the graph of overlapping permutations (since each edge corresponds to exactly one permutation and we know this permutation). At least some of these Hamiltonian cycles (possibly all, which is conjectured), can be extended to u-cycles for permutations via linear extensions of partially ordered sets as described in [CDG1992].

Lemma 1. *Let H_n be the multidigraph with vertex set S_{n-1} in which each $p \in S_n$ is the edge $\text{st}(p_1 \cdots p_{n-1}) \rightarrow \text{st}(p_2 \cdots p_n)$. Then $P(n)$ is the line digraph of H_n , and the cyclic orderings of S_n whose consecutive entries are arcs of $P(n)$ are exactly the Eulerian circuits of H_n . In particular $\text{Ham}(P(n)) \leftrightarrow \text{Eul}(H_n)$ is a bijection, not an inclusion.*

Proof. By definition $p \rightarrow q$ is an arc of $P(n)$ if and only if $\text{st}(p_2 \cdots p_n) = \text{st}(q_1 \cdots q_{n-1})$, that is, if and only if the head of the edge p of H_n equals the tail of the edge q . A cyclic ordering of all of S_n with consecutive entries adjacent in $P(n)$ is therefore a closed walk in H_n using every edge exactly once, and conversely. $\square$

Hence the 2026 phrase “possibly all” and the 2024 quantifier “*any* Hamiltonian cycle in $P(n)$ ” quantify over the same set, and the Eulerian route of [7] is a reparametrisation of the whole search space rather than a construction landing in a distinguished subclass. What is refuted below is the statement as both sources give it.

Provenance of the quotations, and what they do not fix. The paper [2] is not open access and we did not read it; the belief is quoted only through the two citing papers above, whose v1 e-print L^AT_EX sources we read directly, and the quoted spans are exact substrings of those files. We therefore do not claim that Chung, Diaconis and Graham ever asserted the universally quantified statement, still less that they posed it as a mathematical assertion rather than an expectation: [6] reports a belief at second hand, and its own observation that [2] refer to [3] for a construction carrying a guarantee is consistent with less having been claimed there. The one firm anchor is the 2026 parenthetical “possibly all, which is conjectured”. A reader who holds that no universal assertion was ever posed in [2] should read what follows as settling that conjecture of [7] in the negative and as saying nothing about [2]. Neither published journal version of [6, 7] was obtainable, so we cannot exclude a copy-edit having altered the quoted sentences; see Section 4.

Adjacent work. Isaak [4] proves that the Hurlbert–Jackson digraphs $P(n, k)$, whose vertices are the k -permutations of an n -set and whose arcs are the $(k + 1)$ -permutations, are Hamiltonian for $k \leq n - 3$, using restricted Eulerian cycles and a line-digraph relation of the kind used in Lemma 1; that is an existence result on a different digraph over an n -letter alphabet, and it says nothing about acyclicity of the order forced by a cycle of the order-isomorphism digraph $P(n)$. Asplund and Fox [1] work with the digraph $P(n)$ itself, count its short cycles and raise the existence and counting question for Hamiltonian cycles

in it; neither their counts nor that question concerns the order forced on the letters, so they settle nothing about u-cycles. The conjecture quoted above is restated, not resolved, in [7]. We claim no priority for what follows.

2. THE COUNTEREXAMPLE

Theorem 2. *Let $n = 4$, $N = 24$, and let $C = (p_0, \dots, p_{23})$ be*

i	0	1	2	3	4	5	6	7	8	9	10	11
p_i	1234	1243	1432	4312	4132	1324	3142	1423	4123	1342	3421	4213
i	12	13	14	15	16	17	18	19	20	21	22	23
p_i	2134	2341	2413	4231	2314	2143	2431	4321	3214	3241	3412	3124

closed by the arc $p_{23} \rightarrow p_0$, that is $3124 \rightarrow 1234$. Then C is a directed Hamiltonian cycle of $P(4)$ and $D(C)$ contains the directed cycle

$$(1) \quad u_2 < u_5 < u_7 < u_9 < u_{12} < u_{14} < u_{17} < u_{20} < u_{23} < u_2.$$

Hence the implied ordering is not a partial order, C has no linear extension, and C does not extend to a u-cycle for 4-permutations. Consequently the universally quantified statement reported in [6] and conjectured in [7] fails already at $n = 4$.

Proof. Every arc holds. With the rule $p \rightarrow q \iff \text{st}(p_2 p_3 p_4) = \text{st}(q_1 q_2 q_3)$:

0 $\rightarrow$ 1	st(234) = 123 = st(124)	12 $\rightarrow$ 13	st(134) = 123 = st(234)
1 $\rightarrow$ 2	st(243) = 132 = st(143)	13 $\rightarrow$ 14	st(341) = 231 = st(241)
2 $\rightarrow$ 3	st(432) = 321 = st(431)	14 $\rightarrow$ 15	st(413) = 312 = st(423)
3 $\rightarrow$ 4	st(312) = 312 = st(413)	15 $\rightarrow$ 16	st(231) = 231 = st(231)
4 $\rightarrow$ 5	st(132) = 132 = st(132)	16 $\rightarrow$ 17	st(314) = 213 = st(214)
5 $\rightarrow$ 6	st(324) = 213 = st(314)	17 $\rightarrow$ 18	st(143) = 132 = st(243)
6 $\rightarrow$ 7	st(142) = 132 = st(142)	18 $\rightarrow$ 19	st(431) = 321 = st(432)
7 $\rightarrow$ 8	st(423) = 312 = st(412)	19 $\rightarrow$ 20	st(321) = 321 = st(321)
8 $\rightarrow$ 9	st(123) = 123 = st(134)	20 $\rightarrow$ 21	st(214) = 213 = st(324)
9 $\rightarrow$ 10	st(342) = 231 = st(342)	21 $\rightarrow$ 22	st(241) = 231 = st(341)
10 $\rightarrow$ 11	st(421) = 321 = st(421)	22 $\rightarrow$ 23	st(412) = 312 = st(312)
11 $\rightarrow$ 12	st(213) = 213 = st(213)	23 $\rightarrow$ 0	st(124) = 123 = st(123)

The last entry is the wrap arc, so this is a cycle and not a path. No p_i equals p_{i+1} , so neither of the two self-loops of $P(4)$ — which are $1234 \rightarrow 1234$ and $4321 \rightarrow 4321$, and which no Hamiltonian cycle may use — lies on C .

It is Hamiltonian. Group the entries by first letter:

first letter	suffixes occurring
1	234, 243, 432, 324, 423, 342
2	134, 341, 413, 314, 143, 431
3	142, 421, 214, 241, 412, 124
4	312, 132, 123, 213, 231, 321

Each row lists all six permutations of the three remaining letters, and $4 \cdot 6 = 24 = |S_4|$, so the 24 entries are distinct and exhaust S_4 .

The contradiction. Window i occupies the cyclic positions $i, i+1, i+2, i+3$ and is order-isomorphic to p_i , so the rank of u_{i+k-1} inside window i is the k -th letter of p_i . Nine windows suffice, each contributing the inequality between the two letters of consecutive rank shown:

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window i	p_i	positions	ranks	forced
2	1432	(2, 3, 4, 5)	$u_2=1, u_3=4, u_4=3, u_5=2$	$u_2 < u_5$
5	1324	(5, 6, 7, 8)	$u_5=1, u_6=3, u_7=2, u_8=4$	$u_5 < u_7$
6	3142	(6, 7, 8, 9)	$u_6=3, u_7=1, u_8=4, u_9=2$	$u_7 < u_9$
9	1342	(9, 10, 11, 12)	$u_9=1, u_{10}=3, u_{11}=4, u_{12}=2$	$u_9 < u_{12}$
12	2134	(12, 13, 14, 15)	$u_{12}=2, u_{13}=1, u_{14}=3, u_{15}=4$	$u_{12} < u_{14}$
14	2413	(14, 15, 16, 17)	$u_{14}=2, u_{15}=4, u_{16}=1, u_{17}=3$	$u_{14} < u_{17}$
17	2143	(17, 18, 19, 20)	$u_{17}=2, u_{18}=1, u_{19}=4, u_{20}=3$	$u_{17} < u_{20}$
20	3214	(20, 21, 22, 23)	$u_{20}=3, u_{21}=2, u_{22}=1, u_{23}=4$	$u_{20} < u_{23}$
23	3124	(23, 0, 1, 2)	$u_{23}=3, u_0=1, u_1=2, u_2=4$	$u_{23} < u_2$

The last row is a wrap-around window. Concatenating the nine forced inequalities gives (1), a cycle of *strict* inequalities, so no values whatsoever can be assigned to $u_0, \dots, u_{23}$. $\square$

3. THE READING OF THE SOURCE

The wrap-around windows, pinned to published output. Everything above rests on reading all N windows, including the $n-1$ that wrap around; omitting those is the single error that would manufacture a false “acyclic”. That the wrap-around reading is the source’s own is settled by the source’s own worked example at $n=3$ in [arXiv:2408.05984v1](#): for the Hamiltonian cycle

$$132 \rightarrow 312 \rightarrow 123 \rightarrow 231 \rightarrow 321 \rightarrow 213 \rightarrow 132$$

of $P(3)$ they publish the u-cycle word $U_3 = 142342$. Read as a *circular* word of length 6, its six windows of length 3 are

$$(1, 4, 2), (4, 2, 3), (2, 3, 4), (3, 4, 2), (4, 2, 1), (2, 1, 4) \longrightarrow 132, 312, 123, 231, 321, 213,$$

their cycle in order, *including* the two wrap-around windows (4, 2, 1) and (2, 1, 4). Applying the forced-order construction to their cycle returns the transitive reduction $\{a \rightarrow c, a \rightarrow f, c \rightarrow d, f \rightarrow d, d \rightarrow b, d \rightarrow e\}$ on the six positions $a, \dots, f$, which is their published Hasse diagram, and 142342 is one of its linear extensions. By contrast, dropping the wrap-around windows makes the forced order of the cycle of Theorem 2 acyclic — which is exactly the failure mode this control excludes.

Independence of the reading. Theorem 2 does not depend on which of the two natural readings of “the implied ordering” is taken. The source’s literal all-ordered-pairs reading gives 72 forced strict inequalities on the exhibited cycle, with no antisymmetry violation; the consecutive-rank reading used in the proof gives 46. Both contain all nine links of (1).

4. WHAT IS NOT SETTLED

Only $n=4$ is settled. Whether every $n \geq 5$ admits a Hamiltonian cycle of $P(n)$ whose forced order is cyclic is not settled here, and we report no computation at $n \geq 5$. Neither do we report how many Hamiltonian cycles of $P(4)$ have cyclic forced order: no such count is claimed anywhere in this paper. The arc counts 72 and 46 are about the exhibited cycle alone.

Sources not read. The paper [2] itself (no open-access copy exists; publisher access refused), Hurlbert’s 1990 Rutgers thesis [3] (not online, no repository record; it is the source of the existential construction, which this paper does not contradict), and the survey [5] (full text refused at the only location we found). Neither paywalled journal version of [6, 7] was obtainable, so the published-versus-preprint text was never diffed; the risk is low, since [7] independently restates the conjecture as open in 2026.

What is not claimed. U-cycles for n -permutations exist for every n , and every published construction that actually produces one still works; Hurlbert’s construction [3], to which

[2] refer, is untouched. What fails is only the shortcut of starting from an *arbitrary* Hamiltonian cycle. By Lemma 1, “take an Eulerian cycle of the cluster graph and lift it” [7] is not an alternative guarantee: it generates every Hamiltonian cycle of $P(n)$, the one exhibited above included, and [7] itself claims no more for it than that “at least some” of the resulting cycles work.

5. REPRODUCTION AND VERIFICATION

Everything in Theorem 2 is checkable from the printed tables with no computer. The program `verify.py` accompanying this paper reads the 24-entry cycle printed above, together with the published $n = 3$ objects of [6], and re-derives: that the cycle is a Hamiltonian cycle of $P(4)$ using neither self-loop; that each of the nine links of (1) is forced by the window named in the table; that the forced digraph is cyclic under both readings of the source; the arc counts 46 and 72 and the absence of antisymmetry violations; and that Kitaev and Qiu’s own cycle, word $U_3 = 142342$ and Hasse diagram are reproduced by the convention used here, while dropping the wrap-around windows would make the exhibited cycle look acyclic. It reports 73 checks, all passing, in exact integer arithmetic, using only the Python standard library.

The recorded output covers more than this paper claims, and the surplus is not claimed here: a second 24-permutation Hamiltonian cycle of $P(4)$ with cyclic forced order, which this paper does not print; exhaustive verdicts at $n = 3$ and $n = 2$; and a total number of Hamiltonian cycles of $P(4)$ obtained from an integer Matrix–Tree cofactor with published de Bruijn control counts. The program re-derives nothing about $n \geq 5$. The verbatim quotations above are facts about external e-print files and are not checked by the program.

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